The SDN Control Plane

1 Introduction

The **SDN control plane** is the network-wide logic that controls packet forwarding among SDN-enabled devices, as well as the configuration and management of these devices and their services. This represents a shift from per-router control to a logically centralized control model.

2 SDN in Simple Terms

In essence, SDN separates the **"brain"** of the network from the **"brawn."**

A Simple Analogy

- **Traditional Networking:** Like a fleet of taxi drivers where every driver has to know the whole map and decide their own route based on what they see out the window.
- **SDN:** Like **Uber or Google Maps**. There is a central system that sees the whole city's traffic and tells every driver exactly which turns to take. The drivers just follow the instructions.

- **The "Brain" (Control Plane):** Decision-making is moved to a central software called a **Controller**.
- **The "Brawn" (Data Plane):** Switches become "dumb" but fast, focusing only on moving packets based on the Controller's rules.
- **Programmability:** Network management is automated via applications, similar to installing an app on a smartphone.

3 Key Characteristics of SDN

According to Kreutz (2015), an SDN architecture is defined by four fundamental characteristics:

1. Flow-Based Forwarding

Packet forwarding is based on any number of header field values (L2, L3, or L4).

- **Generalized Forwarding:** Contrasts with destination-based IP forwarding.

- **Flow Tables:** Rules are specified in flow tables computed and installed by the control plane.

2. Separation of Data and Control Plane

- **Data Plane:** Consists of relatively simple, fast switches that execute "match plus action" rules.
- **Control Plane:** Consists of software and servers that manage the flow tables.

3. External Network Control Functions

The control software executes on servers distinct and remote from the switches.

- **SDN Controller:** Acts as a "network operating system" maintaining network state.
- **Logical Centralization:** Typically implemented across several servers for scalability and high availability.

4. Programmable Network

The network is programmable via applications running in the control plane.

- **Applications:** Represent the "brains" of the network (e.g., routing, access control, load balancing).
- **APIs:** Use Northbound APIs provided by the controller to manage the data plane.

4 SDN Architecture and Components

The SDN architecture unbundles network functionality into separate layers, much like the evolution from mainframes to personal computers.

Components of SDN

- **SDN-Controlled Switches:** Data plane devices.
- **SDN Controller:** The intermediary managing state and providing APIs.
- **Network-Control Applications:** Logic that determines network behavior.

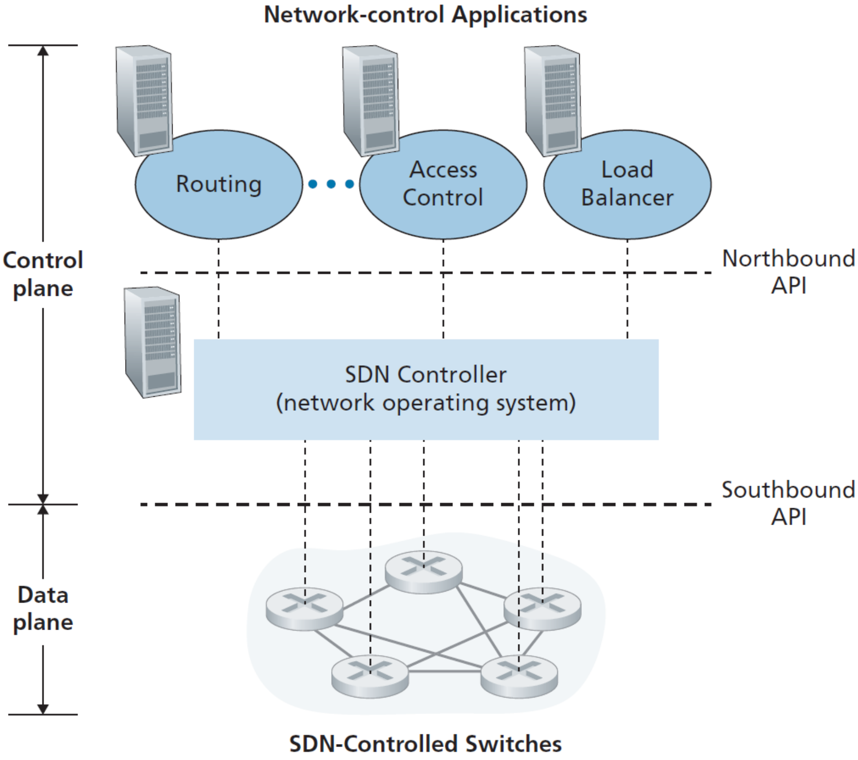

Figure 1: Components of the SDN Architecture: Applications, Controller, and Switches.

Innovation through Unbundling

The separation of hardware, operating systems, and applications in SDN fosters a rich, open ecosystem, enabling rapid innovation across all three layers.